Where Should the Robotaxis Wait? Profit-Driven Fleet Placement

Advanced Data Science (ADS) · End-Term Project Pre-Submission Summary · 23 October 2026

1. Project and group. *Where Should the Robotaxis Wait? Profit-Driven Fleet Placement.* Group ____: Harsh Rohila (240466) · Krish Sharma (240514) · Himank Swami (240474) · Harsh Vardhan (240467)

2. Stakeholder and problem. The fleet operations team of a robotaxi operator such as Waymo, planning a possible New York City launch. Commercial robotaxis are not yet licensed in NYC, so this is launch planning. Riders and the city regulator are secondary stakeholders. With no drivers, the company itself decides where every car waits, so every idle hour and empty mile is a direct cost. Simple rules such as “copy last week” count rides, not money, and cannot react to weather or holidays. *Decisions:* which zones form the launch area; how many cars wait in each zone each hour; whether a car freed in a quiet zone stays or drives empty to a busier one. *KPIs:* total profit (main); profit per car per day; empty miles as a share of all miles; idle car-hours; share of requests served; gain over “copy last week”; service gap between low- and high-demand zones.

3. Research questions and hypotheses. Which launch area earns the most profit for a small fleet? Does planning by expected profit, using a forecast, beat simple rules? When is it worth driving empty to a busier zone?

- **H1:** trip value (fare per minute) differs significantly by zone and hour (tested on real trips).
- **H2:** the data-chosen launch area earns more profit per car than a fixed Manhattan plus Downtown Brooklyn area.
- **H3:** a profit-based placement plan earns more than a ride-count-based plan with the same cars.
- **H4:** a forecast-based plan earns more than “copy last week”.
- **H5:** moving free cars to nearby busy zones earns more than staying, even after paying for the empty miles.

H2–H5 are tested in the fleet simulation on the test months.

4. Data sources (downloaded and audited). Coverage: January–December 2024, 239.5 million trips (19.1–21.3 million per month). January 2025 is used only for the drift check.

Source	Size and coverage	Key variables	Constraints
NYC TLC High-Volume FHV trips (nyc.gov/tlc)	19–21 million trips per month	Request, pickup and drop-off times; zones; miles; trip time; base fare; tips; tolls	Uber/Lyft trips stand in for robotaxi demand; completed trips only; no vehicle IDs
TLC zone lookup and shapefile	263 zones	Borough, airport flag, geometry, neighbouring zones	Zone level only
NOAA NCEI hourly weather (Central Park; LaGuardia and JFK for checks)	99.6–99.8% of hours	Rain, snow, temperature, wind, visibility	Local standard time; one station stands in for the city
US OPM federal holidays	11 per year	Holiday flag	Transcribed by hand

Analysis tables: one row per zone and hour (about 2.3 million rows for 2024) and a zone-to-zone table. 43 observed variables plus derived ones. Waymo’s own trip reports are not usable because location, time and mileage fields are redacted in public versions. Census income and NYC events data are excluded.

5. Integration plan (SSOT). A Spark ETL job cleans trips (one row per trip) and builds two tables: one row per zone and hour (requests, drop-offs, fares, miles, weather, holiday) and one row per zone pair (trip counts, median miles and minutes, used for empty-trip travel). Keys: zone IDs to the zone table (100% matched in a 1-million-trip test); request hour, converted to local standard time, to Central Park weather (99.88% matched); date to holidays; neighbouring zones from the shapefile. Revenue = base passenger fare; tips are excluded because riders may not tip a robotaxi and tips are known only after the trip. Costs = recorded tolls + cost per mile × all miles + idle cost. Empty trips use real-trip median miles and minutes between zones, with straight-line distance for pairs with few trips. No exact duplicates; 0.08–0.09% of trips share a key and are flagged, not dropped. Filters: duration 1 minute to 3 hours, speed 1–80 mph, request before pickup; row counts checked after every join.

6. Methods and BADIR. *Business:* launch-area, placement and wait-or-move decisions; profit KPIs. *Analysis:* hypotheses and baselines (“copy last week”, a ride-count plan, a fixed Manhattan plus Downtown Brooklyn area, cars staying put). *Data:* Spark ETL, SSOT, EDA of trip value by zone and hour. *Insights:* statistical tests and regression on real trips; an optimisation that picks a connected launch area; a gradient-boosting demand forecast with tuning, calibration and SHAP; a fleet simulation (hourly placement, 15-minute matching, three fleet sizes) with plans compared by paired Wilcoxon tests on daily profit. *Recommendations:* placement table and wait-or-move rule; forecasting API in Docker, MLflow tracking, drift monitoring; fairness check.

7. Validation and decision value. Time split: train Jan–Aug 2024, validate Sep–Oct, test Nov–Dec, with holiday days reported separately. January 2025 is the drift check; it includes the congestion-pricing change, so it shows the monitor catching a real shift without claiming anything about the policy. Every plan runs on the same test days, and daily profit is compared in pairs (Wilcoxon). Empty miles and idle time are charged, so no plan wins by moving cars endlessly. Cost per mile is $1.36, tested at $0.74 and $2.00; idle cost is $2 per car-hour, tested at $0 and $5. A minimum service level applies to every zone, and served share and waits are compared for low- vs high-demand zones. Results are associations that hold under the stated assumptions, and each hypothesis is classed as supported, not supported or inconclusive. *Output:* a recommended launch area, fleet-size range, hourly placement table and wait-or-move rule.

8. Limitations and next steps. Uber/Lyft trips are a proxy, and robotaxi riders and prices may differ. Unserved requests are not recorded and there are no vehicle IDs, so fleet results come from simulation. Costs are assumptions, not data. Travel times come from human drivers. Robotaxi constraints (snow, charging, pickup spots, regulation) are simplified. The stakeholder was not interviewed. No deviations are planned. *Next:* ETL and SSOT; EDA and H1; forecast, simulator and launch-area optimisation; H2–H5, deployment and drift monitoring; report and public GitHub repository (semester ends 24 Nov 2026).